Differentiation

Chapter Overview 2

In Year 1 we differentiated terms of the form ax^n. This chapter extends those methods.

1. **Trigonometric, exponential and logarithmic functions**

Find $\frac{d}{dx}\sin x$, $\frac{d}{dx}\ln x$ and $\frac{d}{dx}2^x$.

2. **Chain, product and quotient rules**

Differentiate composite functions, products and quotients, such as $x\sin x$ and e^{2x^2}.

3. **Parametric differentiation**

If $x = \sin t$, $y = e^t$, find the tangent at $(0, 1)$.

4. **Implicit differentiation**

An implicit relationship need not give y explicitly in terms of x.

If $xy + y^2 = 3$, find $\frac{dy}{dx}$.

5. **Rates of change**

A circle's radius increases at 5 cm/s. How fast does its area change when $r = 3$ cm?

Differentiating Trigonometric Functions 3

Differentiate $\sin x$ from first principles.

Recall, as $h \to 0$, $\sin h \approx h$, $\cos h \approx 1 - \frac{1}{2}h^2$.

Also $\sin(x+h) = \sin x \cos h + \cos x \sin h$.

First principles and addition formula

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$
$$= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h}$$
$$= \lim_{h\to 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}.$$

Collect the sine and cosine terms

$$\frac{dy}{dx} = \lim_{h\to 0} \left(\frac{\cos h - 1}{h} \sin x + \frac{\sin h}{h} \cos x \right).$$

Use the small-angle limits

$$\lim_{h\to 0} \frac{\sin h}{h} = 1, \qquad \lim_{h\to 0} \frac{\cos h - 1}{h} = 0.$$

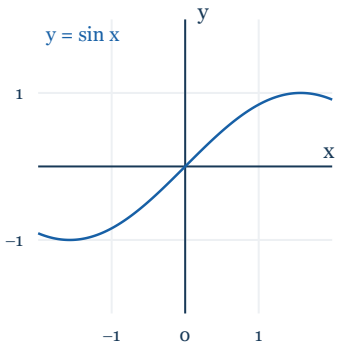
The approximations give $h/h = 1$ and $-h^2/(2h) = -h/2 \to 0$.

$$\frac{dy}{dx} = 0 \sin x + 1 \cos x = \cos x.$$

Why Must We Use Radians? 4

Explanation 1

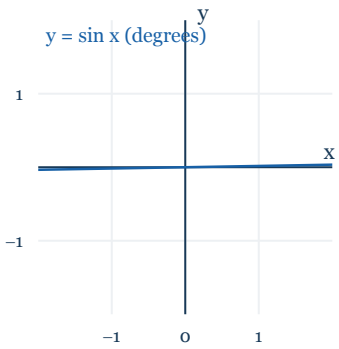
The approximations $\sin x \approx x$ and $\cos x \approx 1 - \frac{1}{2}x^2$, as $x \to 0$, require radians. The first-principles proof used these approximations.



In radians, $\cos 0 = 1$, so the predicted gradient at the origin is 1. This agrees with the graph.

Explanation 2

Compare the graphs with the same scale on both axes.



In degrees, the gradient at the origin is much smaller than 1. The formula $\frac{d}{dx}\sin x = \cos x$ therefore does not apply to degree inputs.

Differentiating Trigonometric Functions 5

Find $\frac{d}{dx}\cos x$.

$$\frac{d}{dx}\cos x = -\sin x.$$

Memorisation tip

Moving right differentiates; moving left integrates.

$\sin x \longrightarrow \cos x$

For the "wrong" function, the sign changes: differentiating $\cos x$, or integrating $\sin x$.

Differentiating Sine and Cosine of kx 6

Find $\frac{d}{dx}\sin kx$.

$k\cos kx$

Find $\frac{d}{dx}\cos kx$.

$-k\sin kx$

These results come from the chain rule, later in this chapter.

Quickfire questions

a. $\frac{d}{dx}\sin 3x$

a

b. $\frac{d}{dx}\cos 5x$

b

c. $\frac{d}{dx}(3\sin 5x)$

c

d. $\frac{d}{dx}(4\cos 3x)$

d

e. $\frac{d}{dx}(-\frac{1}{2}\sin x)$

e

f. $\frac{d}{dx}(-\frac{2}{3}\cos \frac{x}{2})$

f

Example — Stationary Points 7

A curve has equation $y = \frac{1}{2}x - \cos 2x$. Find its stationary points for $0 \leq x \leq \pi$.

Working

Test Your Understanding — Stationary Points 8

A curve has equation $y = \sin 3x + 2x$. Find its stationary points for $0 \leq x \leq \frac{2}{3}\pi$.

Working

Exercise 9A — Page 234

1. a. Given $f(x) = \cos x$, show that

$$f\prime(x) = \lim_{h \to 0} \left(\frac{\cos h - 1}{h} \cos x - \frac{\sin h}{h} \sin x \right).$$

b. Hence prove that $f\prime(x) = -\sin x$.

1a

1b

Exercise 9A — Page 234

7. A curve has equation $y = 2\sin 2x + \cos 2x$. Find its stationary points for $0 \leq x \leq \pi$.

7

8. A curve has equation $y = \sin 5x + \cos 3x$. Find the equation of its tangent at $(\pi, -1)$. **(4 marks)**

8

9. A curve has equation $y = 2x^2 - \sin x$. Show that its normal at $x = \pi$ has equation

$x + (4\pi + 1)y - \pi(8\pi^2 + 2\pi + 1) = 0$.

(7 marks)

9

10. Prove from first principles that the derivative of $\sin x$ is $\cos x$.

You may use the addition formula for $\sin(A+B)$, and that as $h \to 0$,

$$\frac{\sin h}{h} \to 1, \qquad \frac{\cos h - 1}{h} \to 0.$$

10

Differentiating Exponentials and Logarithms 14

Recall from Year 1

$$\frac{d}{dx}e^x = e^x.$$

Just seen

$$\frac{d}{dx}\sin kx = k\cos kx.$$

Applying the same input-scaling idea ⟶

$$\frac{d}{dx}e^{kx} = ke^{kx}.$$

$$\frac{d}{dx}\ln x = \frac{1}{x}$$

$$\frac{d}{dx}a^x = a^x \ln a$$

$$\frac{d}{dx}a^{kx} = ka^{kx} \ln a$$

For positive k, x,

$$\frac{d}{dx}\ln(kx) = \frac{d}{dx}(\ln k + \ln x) = \frac{1}{x}.$$

Scaling the logarithm's input does not change the gradient. Proofs follow using implicit differentiation.

Quickfire Questions — 1/2 15

1. $\frac{d}{dx}(3^x)$

1

2. $\frac{d}{dx}(x^3)$

2

3. $\frac{d}{dx}(\ln(3x))$

3

4. $\frac{d}{dx}(3^{2x})$

4

5. $\frac{d}{dx}(2^{3x})$

5

6. $\frac{d}{dx}(5\ln x)$

6

7. $\frac{d}{dx}(e^{x/2})$

7

8. $\frac{d}{dx}(5\ln(2x))$

8

9. $\frac{d}{dt}(9^t)$

9

10. $\frac{d}{dx}(5(4^x))$

10

11. $\frac{d}{dx}(x^4)$

11

12. $\frac{d}{dx}(\ln(6x))$

12

13. $\frac{d}{dx}(6\ln x)$

13

14. $\frac{d}{dx}(3e^{2x})$

14

15. $\frac{d}{dx}(e^{-x})$

15

Quickfire Questions — 2/2 16

Applied example

$P = 1000(2^t)$ models a rabbit population after t years. When does its rate of increase reach 20,000 rabbits per year?

Application

Test Your Understanding 17

Differentiate $y = (e^x + 2)^2$.

Hint: Expand first.

Derivative

A child's headlice are treated with shampoo. The population after t days is modelled by $P = 460(3^{-2t})$.

a. When are 20 headlice left?

b. Find the rate of change after 3 days.

a

b

Exercise 9B — Pages 236–237 18

4. Find the tangent to $y = 2^x + 2^{-x}$ at $(2, \frac{17}{4})$.

4

5. A curve has equation $y = e^{2x} - \ln x$. Show that the tangent at $x = 1$ is $y = (2e^2 - 1)x - e^2 + 1$.

(6 marks)

5

7. Cambridge's population was 37,000 in 1900 and about 109,000 in 2000. Model the population t years after 1900 by $P = P_0 k^t$.

a. Find P_0 and k.

b. Evaluate dP/dt in 2000.

c. Interpret your answer.

7a

7b

7c

9. Prove that $\frac{d}{dx}a^{kx} = ka^{kx}\ln a$, assuming $\frac{d}{dx}e^{kx} = ke^{kx}$.

9

10. $f(x) = e^{2x} - \ln(x^2) + 4$, $x > 0$.

a. Find $f\prime(x)$.

b. The curve has gradient 2 at $x = a$. Show that $a(e^{2a} - 1) = 1$.

10a

10b

11. Curve C has equation $y = 5\sin 3x + 2\cos 3x$, $-\pi \leq x \leq \pi$.

a. Show that $P(0, 2)$ lies on C.

b. Find the normal at P.

11a

11b

12. Point P lies on $y = 2(3^{4x})$ with $x = 1$. Find the normal at P in the form $y = ax + b$, giving a, b exactly.

12

For Your Interest — Differentiating ln x 24

Prove from first principles that $\frac{d}{dx}\ln x = \frac{1}{x}$, for $x > 0$.

First principles; then the logarithm law

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{\ln(x+h) - \ln x}{h}$$

$$= \lim_{h\to 0}\frac{\ln((x+h)/x)}{h}$$

$$= \lim_{h\to 0}\frac{1}{h}\ln\left(1+\frac{h}{x}\right).$$

Move the multiplier into the power

$$= \lim_{h\to 0}\ln\left(1+\frac{h}{x}\right)^{1/h}.$$

Substitute $n = 1/h$

For the right-hand limit, $n \to \infty$:

$$\lim_{n\to\infty}\ln\left(1+\frac{1/n}{x}\right)^{n}$$

$$= \ln\left(\lim_{n\to\infty}\left(1+\frac{1/x}{n}\right)^{n}\right).$$

Here $(1/n)/x = (1/x)/n$. Continuity of ln lets us move the limit inside.

Using $e^{k} = \lim_{n\to\infty}(1 + k/n)^{n}$, the result is $\ln(e^{1/x}) = 1/x$. The left-hand limit agrees by the same exponential limit.

Hence $\frac{dy}{dx} = \frac{1}{x}$.

Differentiating Combinations of Functions 25

Functions can interact in different ways. Choose the matching rule.

1. **Composite function** $\longrightarrow$ **Chain rule**

$y = f(g(x)) = \sqrt{1 + 3x}$.
Outer function: $f(x) = \sqrt{x}$.
Inner function: $g(x) = 1 + 3x$.

Exercise 9C

2. **Product of two functions** $\longrightarrow$ **Product rule**

$y = f(x)g(x) = x \sin 2x$.

Exercise 9D

3. **Quotient of two functions** $\longrightarrow$ **Quotient rule**

$y = f(x)/g(x) = \ln x/x$.

Exercise 9E

The Chain Rule 26

The chain rule differentiates a composite function: a function within a function.

Example: $y = (3x^4 + x)^5$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$$

Note

The du symbols appear to cancel. This is a memory aid, not a proof.

Full method

Let $u = 3x^4 + x$: this is the inner function.

$$\frac{du}{dx} = 12x^3 + 1.$$

The outer function is $y = u^5$, so

$$\frac{dy}{du} = 5u^4 = 5(3x^4 + x)^4.$$

$$\frac{dy}{dx} = 5(3x^4 + x)^4(12x^3 + 1).$$

Mentally in one go

Differentiate the outer function, keeping its input unchanged, then multiply by the derivative of the inner function.

Think “something to the fifth power”: its outer derivative is five times that same “something” to the fourth power.

Outer derivative: $5(3x^4 + x)^4$.

Now differentiate the inner “something”, $3x^4 + x$.

Inner derivative: $12x^3 + 1$.

Multiply:

$$\frac{dy}{dx} = 5(3x^4 + x)^4(12x^3 + 1).$$

Practice — Chain Rule — 1/2

27

Original function	Outer derivative	Inner derivative
1. $y = (x^2 + 1)^3$	1 outer	1 inner
	Differentiate the cube.	Multiply by the derivative of the inside function.
2. $y = (\ln x)^3$	2 outer	2 inner
	Differentiate the cube.	Multiply by the derivative of the inside function.
3. $y = e^{x^2+x}$	3 outer	3 inner
	Differentiate the exponential.	Multiply by the derivative of the exponent.

Practice — Chain Rule — 2/2

28

Original function	Outer derivative	Inner derivative
4. $y = (2^x + 1)^2$	4 outer	4 inner
	Differentiate the square.	Multiply by the derivative of the inside function.
5. $y = \ln(\sin x)$	5 outer	5 inner
	Differentiate the natural logarithm.	Multiply by the derivative of the inside function.

Further Practice — Rewrite Before Differentiating — 1/2

1. $y = \sin 5x$

1 outer

1 inner

Differentiate the sine.

Differentiate the inside function; multiply the factors.

2. $y = \sin^2 x$

2 outer

2 inner

2 rewrite

Differentiate the square.

Differentiate the inside function; multiply the factors.

Rewrite using brackets first.

3. $y = \sqrt{x+1}$

3 outer

3 inner

3 rewrite

Differentiate the power of one half.

Differentiate the inside function; multiply the factors.

Rewrite using brackets first.

Further Practice — Rewrite Before Differentiating — 2/2

4. $y = \cos^3 2x$

4 outer

4 inner

4 rewrite

Differentiate the cube.

Use the chain rule twice.

Rewrite using brackets first.

5. $y = e^{e^x}$

5 outer

5 inner

Differentiate the exponential.

Differentiate the inside function; multiply the factors.

Test Your Understanding — Chain Rule 31

C3 June 2011, Q1a (2 marks)

Differentiate $\ln(x^2 + 3x + 5)$.

Exam

In general

$$\frac{d}{dx}\ln(f(x)) = \frac{f\prime(x)}{f(x)}.$$

Textbook example

Given $y = \sqrt{5x^2 + 1}$, find dy/dx at $(4, 9)$.

Textbook

Finding dy/dx from dx/dy 32

Sometimes x is given in terms of y, but we want dy/dx.

$$\frac{dy}{dx} = \frac{1}{dx/dy}$$

Proof using the chain rule

$$\frac{dy}{dy} = \frac{dy}{dx}\frac{dx}{dy}.$$

Since $dy/dy = 1$, divide by $dx/dy \neq 0$.

Example 1

Find dy/dx when $x = 2y^2 + y$.

Example 2

Find the gradient of $x = (1 + 2y)^3$ when $y = 1$.

4. Curve C has equation $y = (5 - 2x)^3$. Find its tangent at the point P where $x = 1$.

4

5. Given $y = (1 + \ln 4x)^{3/2}$, find dy/dx at $x = \frac{1}{4}e^3$.

5

9. a. Differentiate $e^y = x$ with respect to y.

b. Hence prove that $y = \ln x$ gives $dy/dx = 1/x$.

9a

9b

10. Curve C has equation $x = 4\cos 2y$.

a. Show that $Q(2, \pi/6)$ lies on C.

b. Show that $dy/dx = -1/(4\sqrt{3})$ at Q.

c. Find the normal at Q, in the form $ax + by + c = 0$, with exact constants.

10a

10b

10c

12. Curve C has equation $y = 4/(2 - 4x)^2$, $x \neq 1/2$. Point A has $x = 3$. Find the normal at A in the form $ax + by + c = 0$, with integer constants.

12

Extension — STEP I 2014, Q4 37

[STEP I 2014 Q4] An accurate clock has an hour hand of length a and a minute hand of length b (where $b > a$), both measured from the pivot at the centre of the clock face. Let x be the distance between the ends of the hands when the angle between the hands is θ, where $0 \leq \theta < \pi$.

Show that the rate of increase of x is greatest when $x = \left(b^2 - a^2\right)^{\frac{1}{2}}$.

In the case when $b = 2a$ and the clock starts at mid-day (with both hands pointing vertically upwards), show that this occurs for the first time a little less than 11 minutes later.

Setup

Maximum

Time

The Product Rule 38

Use the product rule for a product of two functions:

$$y = uv \implies \frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}.$$

Method tips

Write out $u, v, u\prime, v\prime$: the individual derivatives can be complicated. Factorising is typical when finding turning points. Exponentials cannot equal zero.

Memorisation tip

1. Differentiate one factor; leave the other.
2. Do it the other way round.
3. Add. Either order gives the same result.

Example 1

Find dy/dx for $y = x^2 \sin x$.

Write u, v, then differentiate each.

1 factors

Substitute into the product rule.

1 result

Example 2

Find the turning point of $y = xe^{2x}$.

2 derivative

2 x

2 y

Product and Chain Rule Examples 39

Example 3

If $y = e^{4x}\sin^2 3x$, show that

$\dfrac{dy}{dx} = e^{4x}\sin 3x(A\cos 3x + B\sin 3x)$,

finding A, B.

Example 4

Find $f\prime(x)$ for $f(x) = x^2\sqrt{3x-1}$.

Factoring tip

Take out the smallest power, $-1/2$. The other power is one higher, leaving $(3x-1)$. Also take out $1/2$, common to $3/2$ and 2.

The final radical form is optional, but avoids negative powers.

Test Your Understanding — Product Rule 40

Exam-style question 1 (4 marks)

Differentiate $x^2\ln(3x)$, simplifying your answer.

1

Question 2

Given $\frac{dy}{dx} = \frac{1}{6x(x-1)^{1/2}}$, find d^2y/dx^2, simplifying your answer.

2

Examiner comment

“Perhaps the most challenging question in the paper.”

6. Curve C: $y = x^2\cos(x^2)$. Find the tangent at $P(\sqrt{\pi}/2, \pi\sqrt{2}/8)$ in the form $ax + by + c = 0$, with exact constants. **(7 marks)**

6 derivative

6 tangent

7. Given $y = 3x^2(5x - 3)^3$, show that
$\dfrac{dy}{dx} = Ax(5x - 3)^n(Bx + C)$,
finding A, n, B, C.

7

8. Curve C: $y = (x+3)^2 e^{3x}$.

a. Find dy/dx using the product rule.

b. Find the gradient at $x = 2$.

8a

8b

9. a. Differentiate $(2\sin x - 3\cos x)\ln 3x$.

9. b. Differentiate $x^4 e^{7x-3}$.

9a

9b

10. Find dy/dx at $x = 1$ on $y = x^5\sqrt{10x+6}$.

10

The Quotient Rule 46

Use the quotient rule to differentiate a quotient, just as the product rule differentiates a product.

Begin with $y = u/v$.

$$y = uv^{-1}$$

$$\frac{dy}{dx} = -uv^{-2}\frac{dv}{dx} + \frac{du}{dx}v^{-1}$$

$$= v^{-2}\left(v\frac{du}{dx} - u\frac{dv}{dx}\right)$$

$$= \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}.$$

The quotient rule

$$y = \frac{u}{v} \implies \frac{dy}{dx} = \frac{vu\prime - uv\prime}{v^2}.$$

Memorisation tip: “Bottoms first!”

The old denominator v comes first in the numerator; square it in the new denominator.

Note the subtraction in the numerator, unlike the addition in the product rule.

Examples — Quotient Rule

Example 1

Find dy/dx if $y = x/(2x + 5)$.

Set out the factors as for the product rule.

Example 2

Find the stationary point of $y = \sin x/e^{2x}$, $0 < x < \pi$.

Hints for Example 2

When a fraction equals zero, multiply by its nonzero denominator.

Factorise when one side is zero.

A sine/cosine mixture can be converted to tangent by dividing by cosine, checking it is nonzero.

Test Your Understanding — 1/2

Exam-style question 1 (5 marks)

Differentiate $\sin 4x/x^3$, giving your answer in simplest form.

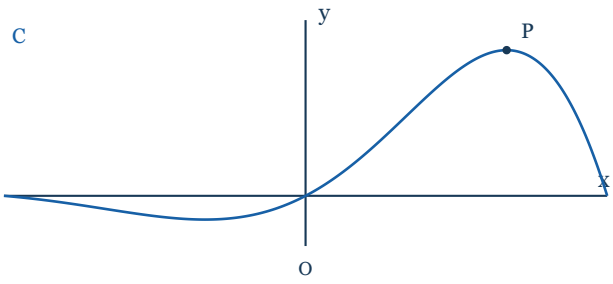
1

Question 2

Curve C: $y = e^{x\sqrt{3}} \sin 3x$, $-\pi/3 \leq x \leq \pi/3$.

a. Find the positive x-coordinate of turning point P, as a multiple of π. **(6 marks)**

b. Find the normal at $x = 0$. **(3 marks)**



2a

2b

6. Find the tangent to $y = e^{x/3}/x$ at $(3, e/3)$.

6

8. Curve C: $x = e^y/(3 + 2y)$.

a. Find its intercept P on the x-axis. **(1 mark)**

b. Find the normal at P in the form $y = mx + c$, with exact constants. **(6 marks)**

8a

8b

9. Differentiate $x^4/\cos 3x$ with respect to x.

9

10. Curve C: $y = e^{2x}/(x-2)^2$, $x \neq 2$.

a. Show that $dy/dx = Ae^{2x}(Bx - C)/(x-2)^3$, finding integers A, B, C.

b. Find the tangent at $x = 1$.

10a

10b

11. $f(x) = \frac{2x}{x+5} + \frac{6x}{x^2+7x+10}$, $x > 0$.

a. Show that $f(x) = 2x/(x+2)$.

b. Hence find $f\prime(3)$.

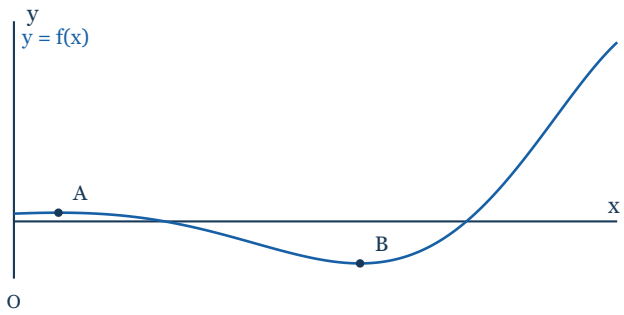
11a

11b

12. $f(x) = 2\cos 2x / e^{2-x}$, $0 < x < \pi$. The curve has a local maximum at A and a local minimum at B.

a. Show that the x-coordinates of A, B satisfy $\tan 2x = 1/2$. **(4 marks)**

b. Find the range of $f(x)$. **(2 marks)**



12a

12b

Extension — Stationary Points 56

STEP I 2010, Q2

The curve $y = \frac{x-a}{x-b}e^x$, with constants a, b, has two stationary points. Show that $a - b < 0$ or $a - b > 4$.

Differentiate

Discriminant

Differentiating Other Trigonometric Functions 57

Differentiate $y = \tan x$.

Use the definition of tangent, then the quotient rule.

$$y = \frac{\sin x}{\cos x}, \quad u = \sin x, \quad v = \cos x.$$
$$u\prime = \cos x, \quad v\prime = -\sin x.$$
$$\frac{dy}{dx} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

More generally

$$\frac{d}{dx}\tan kx = k\sec^2 kx.$$

Tip

This is in the formula booklet, but memorising it saves exam time.

Differentiate $y = \sec x$.

$$y = \frac{1}{\cos x} = (\cos x)^{-1}.$$
$$\frac{dy}{dx} = -(\cos x)^{-2}(-\sin x)$$
$$= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x}\frac{\sin x}{\cos x} = \sec x \tan x.$$

Tip

The quotient rule would work too; the chain rule is often simpler when the numerator is constant.

More generally,

$$\frac{d}{dx}\sec kx = k\sec kx \tan kx.$$

Further Trigonometric Derivatives 58

$$\frac{d}{dx}\tan x = \sec^2 x.$$
$$\frac{d}{dx}\sec x = \sec x \tan x.$$
$$\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x.$$
$$\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot x.$$

Memorisation tip

Adding “co” to the left-hand function adds “co” to the factors in its derivative, and changes the sign.

Example 1

Differentiate $y = \operatorname{cosec} 2x/x^2$.

Example 2

Differentiate $y = \sec^3 x$.

Observation

The power of secant stays the same when differentiating. This will matter in integration.

Test Your Understanding So Far 59

Edexcel C3 June 2013(R), Q5b (3 marks)

Show that $\frac{d}{dx}(\sec^2 3x)$ can be written as $u(\tan 3x + \tan^3 3x)$, where u is constant.

Solution

Using the Reciprocal Derivative 60

When x is given in terms of y, we may want dy/dx in terms of x. Choose an appropriate trigonometric identity:

$\sin^2 x + \cos^2 x \equiv 1, \qquad 1 + \tan^2 x \equiv \sec^2 x.$

Example

Given $x = \tan y$, express dy/dx in terms of x.

Working

Choosing the useful identity

Writing $dy/dx = \cos^2 y$ is correct, but it does not yet express the answer in terms of the known $\tan y = x$.

The identity $1 + \tan^2 y = \sec^2 y$ makes that substitution directly.

Further Examples — Inverse Trigonometric Functions 61

Show that $y = \arcsin x$ gives $dy/dx = 1/\sqrt{1-x^2}$.

Use $\sin^2 y + \cos^2 y = 1$ to express $\cos y$ in terms of $x = \sin y$.

$$y = \arcsin x \implies x = \sin y.$$

$$\frac{dx}{dy} = \cos y, \qquad \frac{dy}{dx} = \frac{1}{\cos y}.$$

For $-1 < x < 1$, the principal arcsine gives $-\pi/2 < y < \pi/2$, so $\cos y > 0$.

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}.$$

Given $y = \arcsin(x^2)$, find dy/dx.

Example

Formula booklet

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}},$$

$$\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}},$$

$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}.$$

Test Your Understanding — Reciprocal and Chain Rules 62

Edexcel C3 January 2011, Q8b,c

Given $x = \sec 2y$:

b. Find dx/dy in terms of y. **(2 marks)**

c. Hence find dy/dx in terms of x. **(4 marks)**

b

c

Textbook

Given $y = \arctan \frac{1-x}{1+x}$, find dy/dx. Use the full chain-rule method.

Textbook

Exercise 9F — Pages 249–251 63

5. Curve C: $y = 1/(\cos x \sin x)$, for $0 < x < \pi$ where defined.

a. Find dy/dx.

b. Determine the number of stationary points.

5a

5b

Exercise 9F — Pages 249–251 64

6. Show that $y = \sec x$ gives $dy/dx = \sec x \tan x$.

7. Show that $y = \cot x$ gives $dy/dx = -\operatorname{cosec}^2 x$.

6

7

STEP I 2011, Q3

Prove the identity

$4\sin\theta\sin(\pi/3-\theta)\sin(\pi/3+\theta)=\sin 3\theta.$ (*)

i. By differentiating (*) or otherwise, show that

$$\cot\frac{\pi}{9}-\cot\frac{2\pi}{9}+\cot\frac{4\pi}{9}=\sqrt{3}.$$

ii. Set $\theta=\pi/6-\phi$ in (*), or use another method, to obtain a similar identity for $\cos 3\theta$. Deduce that

$\cot\theta\cot(\pi/3-\theta)\cot(\pi/3+\theta)=\cot 3\theta.$

Show that

$$\operatorname{cosec}\frac{\pi}{9}-\operatorname{cosec}\frac{5\pi}{9}+\operatorname{cosec}\frac{7\pi}{9}=2\sqrt{3}.$$

Prove $4\sin\theta\sin(\pi/3-\theta)\sin(\pi/3+\theta)=\sin 3\theta.$

Use two factor formulae:

$2\sin A\sin B=\cos(A-B)-\cos(A+B),$

$2\sin A\cos B=\sin(A+B)+\sin(A-B).$

These follow by expanding the right-hand sides using addition formulae and collecting terms.

Proof

Extension Solution — Part 2 of 2 — 1/2 67

i. Differentiate the identity

A product of several factors is differentiated one factor at a time, keeping the others, then adding.

$$(uvwt\cdots)' = u'vwt\cdots + uv'wt\cdots$$
$$+uvwt'\cdots + \cdots .$$

i

Extension Solution — Part 2 of 2 — 2/2 68

ii

Final identity

Parametric Differentiation 69

Parametric equations define x, y (and possibly z) in terms of another parameter, such as t.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad dx/dt \neq 0.$$

Textbook example

Find the gradient when $t = 2$ for $x = t^3 + t$, $y = t^2 + 1$, $t \in \mathbb{R}$.

Gradient

Proof

By the chain rule,

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}.$$

Rearrange to obtain the formula.

Find the normal at $\theta = \pi/6$ on $x = 3\sin\theta$, $y = 5\cos\theta$, $0 \leq \theta < 2\pi$.

Normal

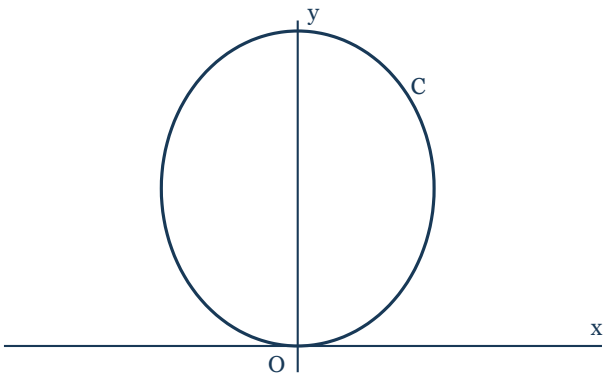
Test Your Understanding — Parametric Tangent 70

C4 June 2012, Q6

Curve C: $x = \sqrt{3}\sin 2t$, $y = 4\cos^2 t$, $0 \leq t \leq \pi$.

a. Show that $dy/dx = k\sqrt{3}\tan 2t$, finding k. **(5 marks)**

b. Find the tangent at $t = \pi/3$, in the form $y = ax + b$. **(4 marks)**



a

b

2. a. Find the tangent to $x = 3 - 2\sin t$, $y = t\cos t$, at $t = \pi$.

2a

2. b. Find the tangent to $x = 9 - t^2$, $y = t^2 + 6t$, at $t = 2$.

2b

3. a. Find the normal to $x = e^t$, $y = e^t + e^{-t}$, at $t = 0$.

3a

3. b. Find the normal to $x = 1 - \cos 2t$, $y = \sin 2t$, at $t = \pi/6$.

3b

4. Find the points of zero gradient on

$$x = \frac{t}{1-t}, \qquad y = \frac{t^2}{1-t}, \quad t \neq 1.$$

You need not classify them as maxima or minima.

4

5. Curve C: $x = e^{2t}$, $y = e^t - 1$, $t \in \mathbb{R}$.

a. Find the tangent at $t = \ln 2$.

b. Show that C has no stationary points.

5a

5b

6. Curve C: $x = (t^2 - 3t - 4)/t$, $y = 2t$, $t > 0$. A tangent l_1 is parallel to $y = x + 5$. Find l_1.

6

7. $x = 2\sin^2 t$, $y = 2\cot t$, $0 < t < \pi/2$.

a. Find dy/dx in terms of t.

b. Find the tangent at $t = \pi/6$.

7a

7b

Exercise 9G — Pages 252–254 77

8. Curve C: $x = 4\sin t$, $y = 2\operatorname{cosec} 2t$, $0 < t < \pi$, where defined. Point $A = (2\sqrt{3}, 4\sqrt{3}/3)$.

a. Find t at A.

b. Show that the normal l at A is $9x + 12y - 34\sqrt{3} = 0$.

8a

8b

Exercise 9G — Pages 252–254 78

9. Curve C: $x = t^2 + t$, $y = t^2 - 10t + 5$, $t \in \mathbb{R}$. At P the gradient is 2.

a. Find P.

b. Find the tangent at P.

c. Show that this tangent does not meet C again.

9a

9b

9c

10. Curve C: $x = 2\sin t$, $y = \sqrt{2}\cos 2t$, $0 < t < \pi$.

a. Find dy/dx in terms of t.

b. Find the normal l at A, where $t = \pi/3$, in the form $ax + by + c = 0$.

c. Prove that l meets the curve only at A.

10a

10b

10c

11. $x = \cos t$, $y = \frac{1}{2}\sin 2t$, $0 \le t < 2\pi$.

a. Find dy/dx.

b. Find the tangent at A, where $t = \pi/6$.

c. Find the two other tangents l_1, l_2 parallel to the tangent at A.

11a

11b

11c

Implicit Differentiation 81

An explicit equation makes y the subject, for example $y = x^2 + 3x$, so we can calculate y directly from x.

An implicit relation may instead be

$x^2 + y^2 = 8x$ or $\cos(x + y) = \sin y$.

Even with $y = x^2$, we differentiate *both* sides with respect to x:

$$\frac{d}{dx}(y) = \frac{d}{dx}(x^2) \quad \Longrightarrow \quad \frac{dy}{dx} = 2x.$$

The derivative of y with respect to x is, by definition, dy/dx.

Two things to remember

1. Differentiate each side of the equation, using the chain rule where needed.
2. Differentiating y with respect to x gives dy/dx.

Examples — Implicit Differentiation 82

Use the chain rule

"Something squared" differentiates to twice that "something", multiplied by its derivative.

For y^2, multiply $2y$ by dy/dx.

1. $\frac{d}{dx}(y^2)$ 1
2. $\frac{d}{dx}(\sin y)$ 2
3. $\frac{d}{dx}(e^y)$ 3
4. $\frac{d}{dx}(xy)$ 4

In general

$$\frac{d}{dx} f(y) = f'(y)\frac{dy}{dx}.$$

5. $\frac{d}{dx}(e^{x^2 y})$ 5
6. $\frac{d}{dx}(\tan(x + y))$ 6
7. $\frac{d}{dx}(x^2 + \cos y)$ 7

Why not just y for the derivative of xy?

Unlike the constant a in ax, y varies with x, so the product rule is needed. This is a very common exam calculation.

Further Examples — Implicit Differentiation 83

Textbook

Find dy/dx in terms of x, y where
$x^3 + x + y^3 + 3y = 6$.

Differentiate both sides: remember that 6 becomes 0.
Then make dy/dx the subject.

Find dy/dx at $(1, 1)$ where $e^{2x} \ln y = x + y - 2$.

Tip

Substitute the point early; there is no need to isolate dy/dx first.

Unlike the Year 1 examples, the gradient here can depend on both x and y.

Test Your Understanding — Implicit Curves — 1/2 84

C4 January 2008, Q5

Curve: $x^3 - 4y^2 = 12xy$.

a. Find the two points where $x = -8$. **(3 marks)**

b. Find the gradient at each. **(6 marks)**

2008 a

2008 b

Test Your Understanding — Implicit Curves — 2/2

C4 June 2014(R), Q3

$x^2 + y^2 + 10x + 2y - 4xy = 10$.

a. Find dy/dx, fully simplified. **(5 marks)**

b. Find y when $dy/dx = 0$. **(5 marks)**

Hint: solve simultaneously with the original equation.

2014 a

2014 b

Exam note

This type of implicit-differentiation question is common.

Exercise 9H — Pages 255–256

1. Write $u = y^n$ and use the chain rule to show that $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$.

1

2. Use the product rule to show that $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$.

2

7. Curve C is described by

$2x^2 + 3y^2 - x + 6xy - 1 = 0$.

Find its tangent at $(1, -2)$ in the form $ax + by + c = 0$, with integer constants.

7

8. Curve C has equation $3^x = y - 2xy$. Find the exact gradient at $(2, -3)$.

8

9. Find the exact gradient of

$\ln(y^2) = \frac{1}{2}x\ln(x-1), \quad x > 1,\ y > 0,$

at the point where $x = 4$.

9

10. Curve C satisfies $\sin x + \cos y = 0.5$, with $-\pi < x < \pi$, $-\pi < y < \pi$.

a. Find dy/dx.

b. Find its stationary points.

10a

10b

Exercise 9H — Pages 255–256

11. Curve C: $ye^{-3x} - 3x = y^2$.

a. Find dy/dx in terms of x, y.

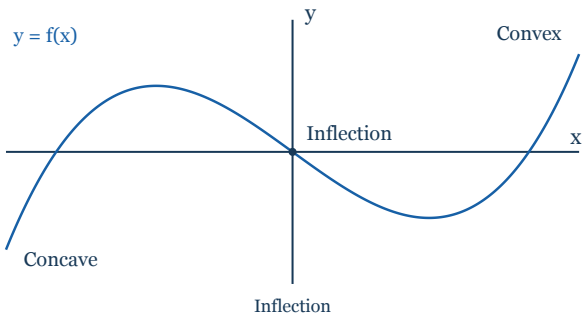
b. Show that the tangent at the origin is $y = 3x$.

11a

11b

Using the Second Derivative

A point of inflection is where concavity changes: concave to convex, or vice versa. Informally, the curve changes the direction in which it is "swerving".



Memorisation tip

Picture a positive cubic, rising overall. Read "concave" then "convex" from left to right: alphabetical order.

Concave

The gradient decreases: $f''(x) \leq 0$ on the interval.

Convex

The gradient increases: $f''(x) \geq 0$ on the interval. The gradient may still be negative while increasing.

Inflection

When f'' is continuous, look for $f''(x) = 0$, then check that its sign changes. The gradient's rate of change is zero there, but the gradient itself need not be zero: an inflection need not be stationary.

Examples — Concavity and Inflection

93

1. Find where $f(x) = x^3 + 4x + 3$ is concave.

1

2. Show that $f(x) = e^{2x} + x^2$ is convex for all real x.

2

3. Find the point of inflection of $y = x^3 - 2x^2 - 4x + 5$.

3

Exercise 9I — Pages 259–261

94

9. Given $y = x(3x - 1)^5$:

a. Find dy/dx and d^2y/dx^2.

b. Find the points of inflection.

9a

9b

Exercise 9I — Pages 259–261

11. A curve has equation $y = \frac{1}{3}x^2 \ln x - 2x + 5$, $x > 0$. Show that it is convex for all $x \geq e^{-3/2}$.

11

Relating Rates of Change

Find the rate of change of the area A of a circle when $r = 3$ cm, given $dr/dt = 5$ cm s^{-1}.

First represent the statements mathematically.

“Rate of change of area”

Area rate

“Radius changes at a rate of 5”

Radius rate

“Area of a circle”

Area

Tip

When you see “rate”, think differentiation with respect to time.

Build the chain rule

Copy the desired numerator and denominator into the diagonal positions, then fill the remaining positions with the intermediate variable r.

$$\frac{dA}{dt} = \frac{dA}{\square} \times \frac{\square}{dt}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

3 — Calculate

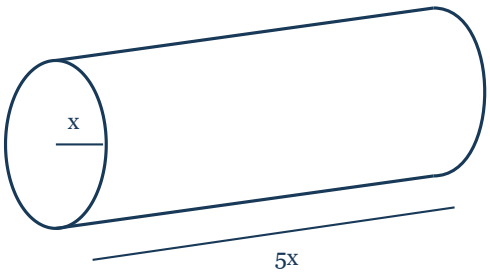
Further Example — Expanding Cylinder 97

Edexcel C4 June 2008, Q3

A right circular cylindrical metal rod expands as it is heated. At time t seconds its radius is x cm and length $5x$ cm. Its cross-sectional area increases at $0.032\ \text{cm}^2\text{s}^{-1}$.

a. Find dx/dt when $x = 2$, to 3 s.f. **(4 marks)**

b. Find the volume's rate of increase when $x = 2$. **(4 marks)**



a

b

It is common for a second part to connect a further pair of rates.

Test Your Understanding — Expanding Cube 98

June 2012, Q2

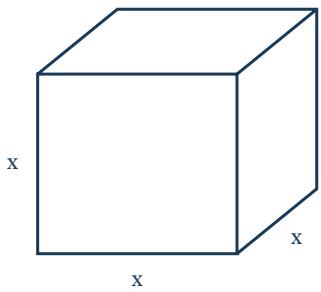
A metal cube expands uniformly as it is heated. At time t seconds, each edge is x cm and the volume is $V\ \text{cm}^3$.

a. Show that $dV/dx = 3x^2$. **(1 mark)**

The volume increases at $0.048\ \text{cm}^3\text{s}^{-1}$.

b. Find dx/dt when $x = 8$. **(2 marks)**

c. Find the surface area's rate of increase when $x = 8$. **(3 marks)**



a

b

c

Tip

$dx/dV = 1/(dV/dx)$.

Exercise 9J — Pages 263–264

Note

Setting up differential equations is covered in Integration. Leave Questions 5, 6, 8, 9, 12 and 14 until that topic has been studied.

1. Given $A = \frac{1}{4}\pi r^2$, $dr/dt = 6$, find dA/dt when $r = 2$.

1

2. Given $y = xe^x$, $dx/dt = 5$, find dy/dt when $x = 2$.

2

Exercise 9J — Pages 263–264

3. Given $r = 1 + 3\cos\theta$, $d\theta/dt = 3$, find dr/dt when $\theta = \pi/6$.

3

4. Given $V = \frac{1}{3}\pi r^3$, $dV/dt = 8$, find dr/dt when $r = 3$.

4

Extension — Cone Optimisation 101

STEP I 2009, Q5

A right circular cone has radius r, height h, slant height l, volume V and curved surface area A:

$$V = \frac{1}{3}\pi r^2 h, \qquad A = \pi r l.$$

i. With A fixed, r is chosen so that V is stationary. Show that $A^2 = 3\pi^2 r^4$ and $l = \sqrt{3}r$.

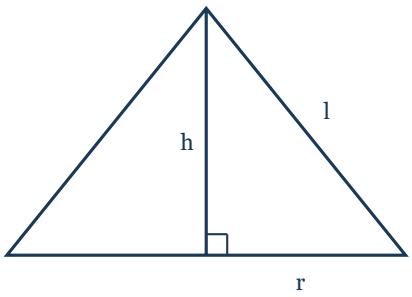
ii. Instead, V is fixed and r is chosen so that A is stationary. Find h in terms of r.

ii

Extension Solution — Fixed Surface Area 102

Solution to i

With A fixed, $l = A/(\pi r)$. A side view of the cone gives $l^2 = h^2 + r^2$.



Geometry

Stationary volume